

Week 6 Major Outcome and Problem Memo: Credit Card Fraud Detection

To: Instructor, Big Data Analytics and Modeling

From: Project 1 Team (8)

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Subject: Week 6 Clinic Submission – Pipeline Status, Preliminary Results, and Corrective Actions

1. Working Preprocessing Pipeline

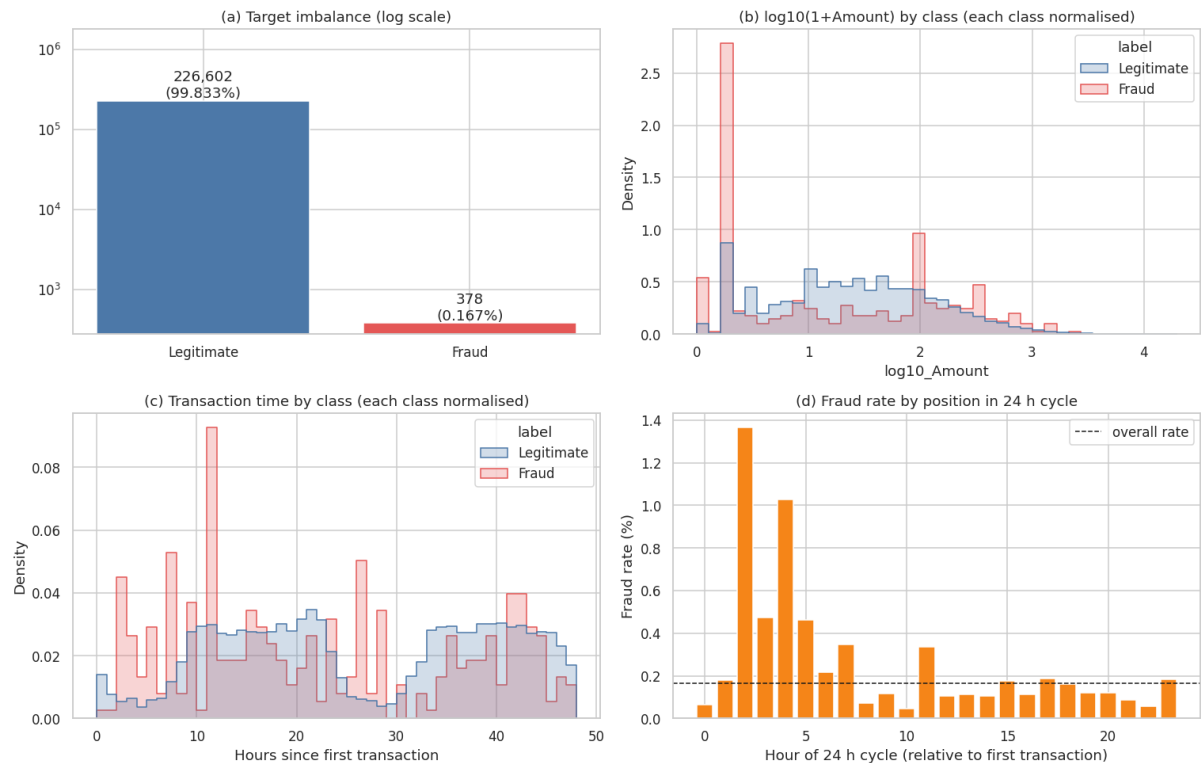


Figure 1 - Exploratory data analysis showing extreme target imbalance and right-skewed Amount distributions necessitating scaling.

The team has established a fully reproducible, leakage-safe pipeline using the training data only.

- Exact duplicate rows (1,081) were removed prior to splitting to prevent identical transactions from appearing in both the training and test sets, which would artificially inflate scores.
- The data was split 80/20 using a stratified split to maintain the 0.167% fraud prevalence in both the training and testing sets.
- Because raw transaction time (seconds elapsed) does not generalize to a new day, it was replaced with cyclic encodings (Time_sin, Time_cos).
- A RobustScaler was applied to the heavily skewed Amount column.
- Every learned step, including scaling and resampling, was placed inside an imblearn pipeline so that it is strictly re-fitted inside each cross-validation fold, preventing validation leakage.

2. Baseline and Preliminary Models

The team trained a baseline and three candidate models using fixed hyperparameters.

- **Naive Benchmark:** A DummyClassifier predicting the majority class (legitimate) to establish a floor.
- **Interpretable Model:** Logistic Regression utilizing SMOTE (Synthetic Minority Over-sampling Technique) to oversample the minority class to 10% of the majority.

- **Stronger Alternatives:** A Random Forest Classifier using `class_weight="balanced_subsample"` and an XGBoost Classifier using `scale_pos_weight` to handle imbalance natively.

3. Metric Comparison Table

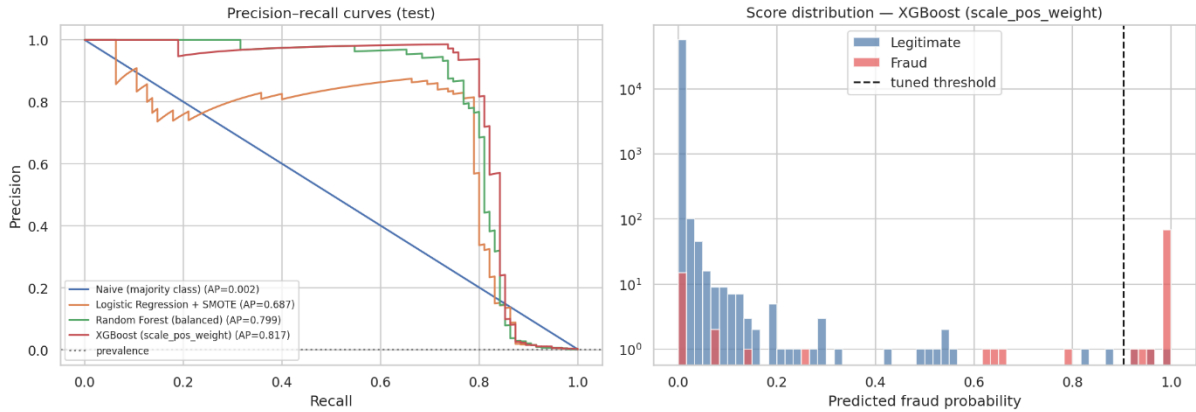


Figure 2 - Precision-Recall curves across all models and the bimodal score distribution for the selected XGBoost model.

Because standard accuracy is misleading (the naive model achieves 99.83% accuracy while catching zero frauds), the team utilized a 5-fold stratified cross-validation strategy focused on Precision-Recall Area Under Curve (PR-AUC). Below are the final evaluation metrics calculated on the strictly held-out test set:

Model	Tuned Threshold	Precision	Recall	F1-Score	PR-AUC
Naive Benchmark	-	0.000	0.000	0.000	0.002
Logistic Regression + SMOTE	0.97	0.830	0.768	0.798	0.687
Random Forest (Balanced)	0.54	0.932	0.726	0.817	0.799
XGBoost (scale_pos_weight)	0.90	0.960	0.758	0.847	0.817

XGBoost was selected as the leading candidate because it achieved the highest cross-validated PR-AUC (0.847 ± 0.035) before any test set evaluation occurred.

4. Major Data/Modeling Problems and Corrective Actions

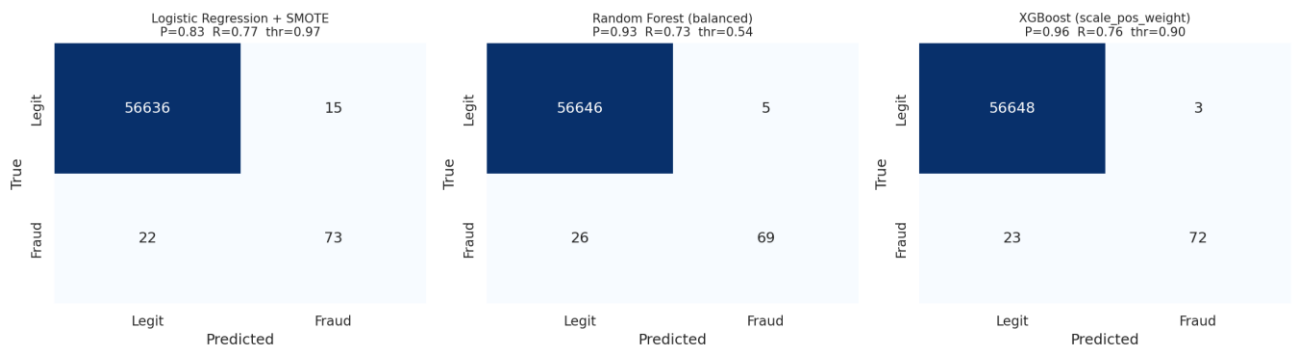


Figure 3 - Confusion matrices illustrating the stark difference in False Positives between Logistic Regression (15) and XGBoost (3).

Problem A: Arbitrary thresholds failing under heavy imbalance.

- **The Issue:** The default 0.5 classification cut-off is arbitrary and ineffective when the target fraud class represents only 0.167% of the training data.
- **Corrective Action:** The team dynamically tuned the decision threshold by maximizing the F1 score using out-of-fold predictions on the training set. This resulted in optimal thresholds far from 0.5, specifically pushing to 0.90 for XGBoost and 0.97 for Logistic Regression.

Problem B: High false positive rates from synthetic oversampling.

- **The Issue:** While the Logistic Regression model with SMOTE successfully caught 73 frauds on the test set, it suffered from lower precision and generated 15 false alarms. In a real banking context, blocking these legitimate customer transactions causes severe dissatisfaction.
- **Corrective Action:** The team successfully reduced false alarms by prioritizing the XGBoost model, which handled the imbalance algorithmically via `scale_pos_weight` instead of generating synthetic data with SMOTE. This dropped the false alarms to 3 while maintaining strong recall by catching 72 frauds.